



Hewlett Packard
Enterprise

Agentic Bug Triage & Routing — Team Contributions

Phase 1: Collaborative Design (Entire Team)

All major architectural decisions were made collectively by the team.

Key Design Contributions:

- Six-Layer System Architecture (Frontend → API Gateway → Kafka → Orchestrator → Agents → External Systems)
- Four-Phase Agentic Triage Pipeline (Context Fetch → Cross-System Correlation → Knowledge Enrichment → AI Synthesis)
- Connector Architecture & Adapter Pattern (JIRA, GitHub, Bugzilla, Confluence, Customer Portal)
- Data Model & Database Design (source_registry, pipeline_context, audit_log, bug grouping)
- Event-Driven Processing & Crash Recovery (Kafka, checkpoint-resume workflow, fault tolerance)
- Redis Caching & WebSocket Streaming Strategy (live bug dashboard, progressive panel updates)
- Security & Data Sovereignty Architecture (JWT authentication, read-only external access)
- Sequence Diagrams, API Contracts & System Workflows

Phase 2: Implementation Contributions

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Backend & Infrastructure

- **Pulkit Jain:** API Gateway, Authentication & Authorization, REST APIs, Service Layer, WebSocket Communication
- **Shivansh Gaur:** PostgreSQL Database Layer, Kafka Event Processing, Redis Caching, Docker & Infrastructure Setup

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AI & Orchestration

- **Disha Jain:** Context Fetch Agent, AI Synthesis Agent, LLM Integration, Pipeline Orchestration
- **Anuj Modani:** Cross-System Fetch Agent, Similarity Search, Knowledge Enrichment Agent, Related Issue Correlation

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External Integrations

- **Om Jain:** JIRA Integration, GitHub Integration, Bugzilla Integration, Confluence Integration, Connector Registry & Adapter Pattern